Chapter 2

→ Ethics in Machine Learning.

Step 1: What is the system doing?

→ Prediction?

→ Classification?

→ Decision-making in high-stakes areas?

(Policing, loans, hiring, healthcare)

→ The system is used for automated decision-making in a high-stakes context.

*** Step 2 → Bias Check

→ Historical Bias

→ Trained on past data reflecting social inequality. (Policing, hiring, loans)

→ Statistical Bias

→ Unbalanced/non-representative dataset.

→ Cognitive Bias

→ Human assumptions influencing design or interpretation.

→ The System suffers from historical bias due to biased training data.

→ if minorities/poor/neighborhoods mentioned → Historical Bias.

Step 3: Transparecy / Black Box

→ Can users explain why the model made this decision?

→ Is it a deep learning / proprietary model?

→ Lack of transparency makes the decision-making process difficult to explain and trust.

Step 4: Accountability

→ who is responsible for wrong decision?

→ Human or AI?

→ Absence of clear accountability raises ethical concerns.

Step 5: Impact / Harm

→ Discrimination?

→ Unfair treatment?

→ Life-altering consequences?

→ The system may lead to unfair and discriminatory outcomes.

Concerns: → algorithmic bias
→ lack of transparency
→ unclear accountability
→ risk of discriminatory outcomes.

How system can be improved?

→ Use more diverse and representative data.

→ Regularly audit the model for Bias.

→ Introduce human in the loop decision making.

→ Improve model interpretability.

→ Ensure transparency and documentation.

→ Bias can be reduced through better data, regular audits, and explainable AI techniques.

→ Balance Between AI and Human assessment.

↳ Informed Consent

↳ Collaboration

↳ AI as support

↳ Trust in Expertise

↳ User Autonomy

↳ Second Opinion

↳ Improve AI transparency.

AI informs decision, humans make decisions.

→ Ethics in social media and research:-

Key concerns →

PIAREHD

P → Public vs Private → Privacy Concern

I → Informed consent → use data without permission

A → Anonymity → identifable details raises concerns about reidentification of individuals.

R → Risk of Harm → concern of harm health/psychological issue.

E → Ethical framework → Concern of violating ethical framework.

H → Human Vulnerability → Concern of vulnerable group.

D → Data Ownership → concern of misuse and loss user contrl over persnl information.

Scan Question: Detection:-

Step 1 → Nature of Data

→ Is the data sensitive?

(mental health, abuse, trauma, personal life)

→ If sensitive → high ethical risk.

→ The study involves sensitive personal data, increasing ethical risk.

Step 2 → Public vs private

→ was the data from a forum, group or a personal profile?

→ Did users have a resonable expectation of privacy

→ Accessible ≠ Ethical

→ Although the data was publicly accessible, participants had a reasonable expectation of privacy.

Step 3 → Informed Consent

→ were participants clearly informed?

→ was consent explicitly obtained (not just T&C)?

→ 90% time answer is → No proper consent

→ The reashercers did not obtain informed consent, which is a major ethical violation.

Step 4 → Anonymity

→ were names, usernames, images or profiles included?

→ Is re-identification possible?

→ violation = Lack of anonymization

→ Personal and identifiable information was included without anonymization, increasing the risk of re-identification.

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Step 5: Risk of Harm:

→ Possible psychological, social, or reputational harm?

→ Are participants vulnerable?

→ This exposes participants to potential ~~and~~ psychological and social harm, especially as they belong to a vulnerable group.

☑ How it could done ethically?

→ Treat the data as private.

→ Obtain informed consent before data use.

→ Anonymize all identifiable information.

→ Minimize sensitive details in reporting.

→ Allow participants the right to withdraw.

→ All process should be transparent.

Consent form

Dear user,

we value your privacy and want you to clearly understand how we collect and use your data. please read the information below before agreeing the terms and conditions.

1.

2.

3.

You have the right to:

i. Access or delete your data.

ii. Withdraw consent at any time.

By signing below, you consent the collection and use of your data as described ~~below~~ above.

Name:

Date:

Signature:

☐ I agree to the terms above.

☐ I prefer not to agree in data collection.

Chapter 4

Ethical Concern in Data collection:

1. What Ethical principle was violated?
2. What should proper practice look like?
3. What harm occurred / could occur?

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Step 1: Consent check

→ Was consent clear and detailed?

→ or not?

→ Vague Consent ≠ informed consent.

→ The principle of informed consent was violated.

Step 2: What should informed consent include?

→ Purpose of data collection

→ Types of data collected

→ How data will be stored

→ Who will have access

→ Duration of storage

→ Right to withdraw.

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Informed consent
privacy
confidentiality

STep 3: Data Type check:

→ Is the data personal?

→ Is the health-related?

If yes → hight ethical sensitivity

→ Health and identifiable data require stricter ethical protection.

Step 4: Confidentiality & Privacy

→ was identifiable data leaked or shared

→ Name, heart rate, location?

If yes → violation.

→ confidentiality and privacy were compromised due to exposure of identifiable data.

Step 5: Harm / Risk

→ Psychological harm?

→ Discrimination?

→ Loss of trust?

→ Such data leakage may cause psychological, social or reputational harm.

How could be done ethically?

→ Provide clear and detailed informed consent.

→ Minimize data collection

→ Anonymize or psedonymize data

→ Secure storage and access control.

→ Share only aggregated data.

Ethical Issues to consider →

→ Informed consent

→ Data Minimization

→ Transparency

→ Right to withdraw

→ Data Security

ICAS

I → Informed consent

C → Confidentiality

A → Anonymity

S → Storage & Sharing.

Additional →

HIPAA → Health Insurance Portability and Accountability (HIPAA) (1996)

→ Designed to Protect personal health information

Updated in 2013 →

→ Patients can access eletronic records and genetic data.

GDPR → General Data Protection Regulation.

Data Protection only allowed if →

→ the individuals has given explicit consent or

→ it is required to fullfill a contract or

→ it is necessary for legal obligations or

→ it protects vital interests or serves public interest or official authority

GDPR Emphasizes:

→ User control

→ transparency

→ accountability.

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